

Hero inverted-U — three model layers (v1)

Date: 2026-07-20

Audience: Charles + Alex read-through

Canonical hero folder: `sports/datasets/mbb/exports_inverted_u_v0/alex_side_by_side_v0/`

Related: `05_Model_Nesting_Note_v1.md`, `30_Alex_Gates_inform_status_outline.md`, `../3-Master_Plan/BINDING_Selection_is_its_own_step.md`

0. BINDING — selection is its own step (Charles #1 confusion)

Do not merge “describe the peer environment” with “who wins the scarce slot.”

Mechanism	Object	Question
Environment	<code>L_net = B(.) - D(.)</code>	How do peers help vs hurt (development, visibility, crowding around you)?
Selection	Alex <code>S_i = A_i - λ · L_C</code>	Who gets drafted? Congestion enters the advancement rule .

- **Hero (Layer A):** outcome only — draft rate vs pool quality. Does **not** identify environment vs selection channel.
- **538D (Layer C):** makes **selection explicit** — rank by score → top **K**; knockout = talent-only vs congestion-in-score.

The hero describes outcomes; Alex’s equation describes who gets selected; the sim tests congestion in the **selection rule**.

1. Locked hero estimand (fixed target for all tests)

All Layer A/C comparisons use **this** empirical spec unless explicitly noted as robustness.

Field	Value
Outcome	<code>Y_draft</code> (ever drafted), mean rate per bin
X-axis	<code>poolq_loo</code> (LOO mean teammate <code>perf</code>)
perf	ESPN box ppm, z-scored within collegiate season
Bins	16 quantile (<code>poolq_binning='quantile'</code>)
Winsor	<code>poolq_loo</code> at (0.01, 0.99) before binning
min_minutes	20 (applied before LOO when rebuilding from box)
Panel source	<code>use_prebuilt_panel_csv=False</code> for canonical rebuild path
Draftee filter	Off (<code>restrict_teams_by_draftees=False</code>)
Seasons	2011–2021
Sample (Jul 2026 lock)	62,180 player-seasons; 1,134 drafted

File naming slug: `empirical_ppm_poolq_loo_16quantile_winsor0199_min20_2011`

Shape (qualitative): rise through middle ventiles → plateau bins 12–15 (~2.3–2.6%) → **dip at bin 16 only** (~1.2%).

2. Three layers — do not merge into one “model”

Layer	Object	Repo anchor	What a “pass” proves	What it does not prove
A. Phenomenological	$Y \sim \beta_0 + \beta_1 \cdot \text{poolq_loo} + \beta_2 \cdot \text{poolq_loo}^2$	530 / <code>panel_build.draft_poolq_quadratic_coeffs</code>	Inverted curvature in the real panel on the hero estimand	Mechanism; that NBA uses this score
B. Structural (theory)	Two parts: (1) <code>L_net = B(.) - D(.)</code> environment; (2) Alex <code>S_i = A_i - \lambda \cdot L_C</code> selection rule	Nesting note §2–3	Environment help/hurt and D enters selection (separate step)	Separate estimation of B(Q) and D(Q) in v1; hero ≠ selection equation
C. Generative (sim)	Soft assign → score → top-K	538D CELL 10–12, <code>tier1_generative_eda.py</code>	Congestion in the score changes selection curves; talent-only fails	Bin-for-bin reproduction of empirical <code>poolq_loo</code> ventiles

Alex side-by-side (v1): Layer A hero PNG + Layer C sim PNG on same Y (advancement rate), honest X (empirical LOO quality vs sim LOO bins), plus limitation sentence below.

3. How each layer is tested

Layer A — regression on real rows

- Filter panel to hero estimand → OLS `Y_draft ~ 1 + poolq_loo + poolq_sq`.
- **Pass:** `β2 < 0` (concave); optional overlay tracks ventile bin means.
- **Not required:** R², causal interpretation, or match to sim.

Layer B — prose + ingredient knockouts

- No single estimated formula in v1.
- **Part 1:** B – D frames the **environment** (help vs hurt among peers).
- **Part 2:** Alex score frames **selection** — who gets the slot — as its **own step**; congestion enters the **selection rule**, not “the whole model.”
- **Pass:** Narrative consistency + sim shows λ=0 (ability-only **selection**) **fails** the generative congestion story.

Layer C — simulation

- Draw synthetic league → assign rosters → rank by score → select top **K** → bin by `poolq_loo` (16 quantile).
- **Knockout A (talent-only)**: `score_mode='ability'` → **monotone** selection rate on `poolq_loo` (bin 1 ~0.7% → bin 16 ~28%); no elite dip.
- **Knockout B (congestion)**: `score_mode='loo_gap_plus_ability'`, `loo_pool_l_mode='crowding_smooth'`, `w=0.5` → elite-tail **compression** (bin 16 ~16% vs ~28% talent-only); not pointwise hero match.
- **Script**: `sports/scripts/hero_model_reset_bundle.py` writes CSV + side-by-side PNG.

4. Simplification strategies (both are valid)

Strategy	Use for	Procedure
Bottom-up	Layer A	Start quadratic → decompose Q only at Rung 2.5 (<code>poolq_loo</code> vs <code>crowding_smooth</code> in CELL 5d)
Top-down	Layer C	Full generative stack → zero λ / ability-only / drop assignment knobs → record what breaks

Use **bottom-up first** for the hero; **top-down** for 538D mechanism strip-down.

5. Simulation plumbing vs core claim

Knob	Role	v1 gate?
λ / w on L_C in score	Core mechanism	Yes
Top-K selection	Scarcity	Yes (conceptual)
Soft assignment (τ , T_j)	Realistic pools	Helpful, not theorem
Assortative sort-and-chop overlay	Diagnostic (Plot A)	No
Faithful-538 empirical team sweeps	Realism stretch	No — see §6
Bin-for-bin LOO match to hero	North-star R&D	No — see §6

Path II (locked): Generative figure may bend on LOO `poolq_loo` bins (CELL 12 default) while empirical hero is the replicated fact on the same conditioning object — still **not** claiming pointwise match without a dedicated calibration pass.

6. Explicitly deferred (stretch, not v1 success criteria)

Do **not** block Alex draft or side-by-side on:

- 1. Bin-for-bin replication of hero ventile draft rates from sim on `poolq_loo`.
- 2. Rivanna / faithful_538 full sweeps (`outputs/simulation_sweeps/faithful_538_sweep*`) until minimal λ + K story is documented.
- 3. Separate B(Q) and D(Q) estimation on one axis.
- 4. Assortativity grids, empirical team cloning, HS binning, multiplicative rewrite.

These remain **parallel R&D** after the minimal POC bundle is frozen.

7. Limitation sentence (Alex slide / manuscript §5)

*The empirical stylized fact is defined on **leave-one-out pool quality** among teammates; the minimal generative proof-of-concept selects on a **score that penalizes leave-one-out viable-peer congestion** and plots binned **selection rates on the same LOO quality axis**. We do **not** claim bin-for-bin reproduction of the empirical ventile draft rates in v1; we claim that **talent-only selection is insufficient** and that **congestion in the score** can bend advancement curves in a disciplined artificial league.*

(Source: nesting note §4; shortened for slides.)

8. Keepers vs reorder (Jul 2026 reset)

Keep: `530` pipeline, hero triplet, nesting note, 538D CELL 10–12, Pertinent Thoughts sensitivities, `alex_side_by_side_v0/` exports.

Reorder claims: Phenomenon first (hero) → minimal sim ingredient (λ) → theory prose — not “one formula does everything.”

Deliverables from this reset: this memo, `lpm_hero_coefficients.txt`, generative knockout CSVs, `inverted_u_side_by_side_empirical_vs_generative.png` in `alex_side_by_side_v0/`.